Censoring in Survival Analysis

Right, left and interval censoring; Type I, II and III designs;
random versus informative censoring

Study notes with biological examples and illustrations

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1 Type I censoring (fixed-time censoring)

Design. The time at which the study starts and stops is fixed in advance. All subjects enter at the same time and are followed for the same period c. Those still alive at the end are censored; their survival times are recorded as at least the length of the study period. Animals lost or accidentally killed have survival times from the start of the experiment to the time of loss or death. Subjects that die during the study have exact survival times, from the start to the time of death.

Random quantity: the number of observed events. The censoring time is fixed.

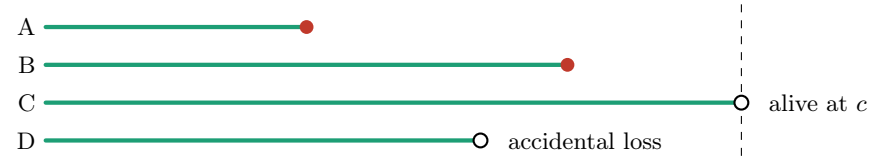
Likelihood:

$$L = \prod_{i=1}^{n} f(t_i)^{\delta_i} \, S(c)^{1-\delta_i}.$$

Biological examples.

- *Toxicology:* 50 mice receive a carcinogen and are observed for exactly 24 weeks. Mice dying earlier have exact times; the rest are censored at week 24.
- *Ecotoxicology:* zebrafish embryos or *Daphnia* exposed to a pesticide for a fixed 96 hours.
- *Vaccine challenge:* animals challenged with a pathogen and observed for a fixed 28 days.
- *Reliability analogue:* a life test of 100 devices run for exactly 1 000 hours.

Type I: fixed stop time, simultaneous entry



Study ends at fixed time c. The number of deaths is random.

2 Random (non-informative) versus informative censoring

Non-informative (random) censoring assumes the censoring mechanism is unrelated to the underlying event time. Standard methods (Kaplan–Meier, log-rank test, Cox regression) assume this.

Informative censoring means censoring is related to the event time, which requires special methods of analysis.

- *Non-informative example:* a participant relocates for a new job. This is unrelated to prognosis, so standard methods remain valid.
- *Informative example:* patients in a chemotherapy trial withdraw because side effects are making them sicker. Censoring is then related to the event time, and Kaplan–Meier will overestimate survival.
- *Remedies:* sensitivity analyses, competing-risks methods, or inverse-probability-of-censoring weighting.